Large Language Models for the
Economic and Social Sciences:

AI-Augmented Surveys

Indira Sen, Georg Ahnert, HWS 2026
https://github.com/dess-mannheim/LLM4ESS

Summary of Last Week and Today

Last session we learned:

- What is content analysis
- What do LLMs have to with it?
- Can LLMs really help with content analysis?
- Project pitches

Today:

- AI for survey development
- In Silico Samples
- Behavioral Experiments
- Validation and Pitfalls

Recap: What is content analysis?

Why?

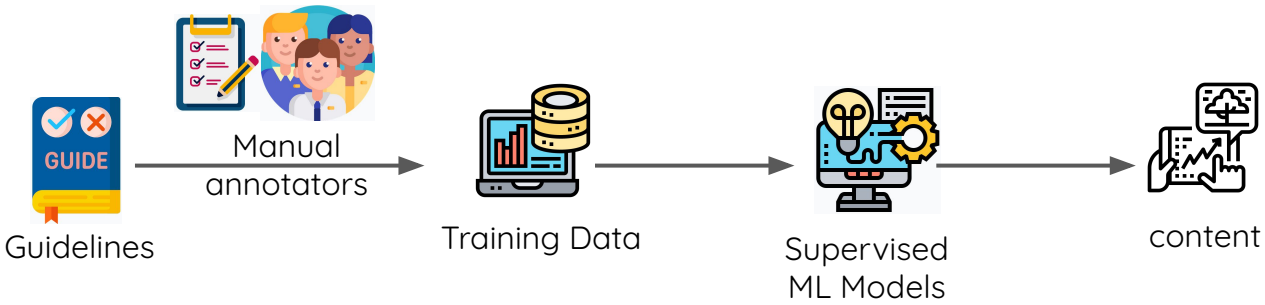
- Content analysis:
  - “any technique for making inferences by objectively and systematically identifying specified characteristics of messages” [Holsti, 1969]
  - “a systematic, replicable technique for compressing many words of text into fewer content categories based on explicit rules of coding” [Berelson, 1952; GAO, 1996; Krippendorff, 1980; Weber, 1990]
- Questions content analysis can help us answer:
  - How does news media represent the immigration crisis?
    NYtimes articles frames [Mendelsohn, 2021]
  - What are topics that lead to arguments in long-term relationships?
    r/relationship_advice posts topics
  - How do citizens perceive the performance of politicians during the pandemic?
    Twitter posts stance

Before LLMs: ‘Modern Pipeline’

- unsupervised



- supervised



Before LLMs: ‘Modern Pipeline’

- unsupervised

Unsupervised
ML Models

Qualitative
assessment

content

- supervised



Guidelines



Manual
annotators



Training Data



Supervised
ML Models



content

Before LLMs: ‘Modern Pipeline’

- unsupervised

Unsupervised ML Models

Qualitative assessment

content

- supervised



Guidelines



Manual annotators

crowdworkers

amazon mechanical turk

Prolific

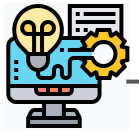
OR

Research assistants





Training Data



Supervised ML Models



content

Before LLMs: ‘Modern Pipeline’

- unsupervised

Unsupervised ML Models

Qualitative assessment

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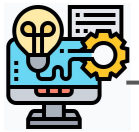
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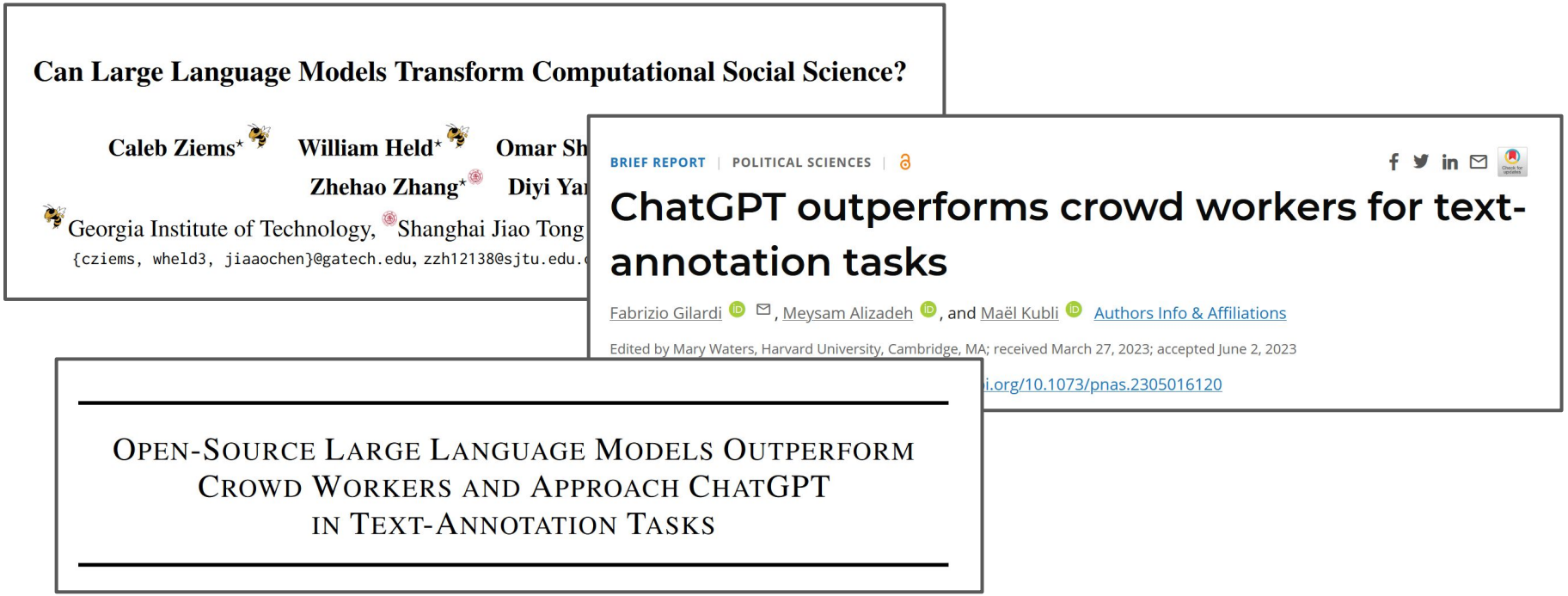
Supervised ML Models



content

- report agreement b/w annotators
- throw out instance with disagreement

After LLMs



Can Large Language Models Transform Computational Social Science?

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BRIEF REPORT | POLITICAL SCIENCES |

ChatGPT outperforms crowd workers for text-annotation tasks

Fabrizio Gilardi, Meysam Alizadeh, and Maël Kubli Authors Info & Affiliations

Edited by Mary Waters, Harvard University, Cambridge, MA; received March 27, 2023; accepted June 2, 2023

i.org/10.1073/pnas.2305016120

OPEN-SOURCE LARGE LANGUAGE MODELS OUTPERFORM CROWD WORKERS AND APPROACH CHATGPT IN TEXT-ANNOTATION TASKS

After LLMs

Model / Data	Baselines		FLAN-T5					FLAN	Chat	text-001				text-002	text-003
	Rand	Finetune	Small	Base	Large	XL	XXL	UL2	ChatGPT	Ada	Babb.	Curie	Dav.	Davinci	Davinci
Utterance Level Tasks															
Dialect	4.5	41.5	1.9	2.3	15.8	16.5	22.6	23.7	15.0	5.3	5.6	6.0	10.9	10.5	16.9
Emotion	16.7	91.7	23.9	65.3	69.1	65.9	66.7	70.3	46.2	44.6	16.1	18.7	19.3	39.8	36.5
Figurative	25.0	94.4	23.6	29.0	25.4	40.2	56.0	64.0	50.2	25.0	24.4	25.0	28.8	52.0	60.6
Humor	50.0	73.1	52.0	51.8	56.2	59.0	50.6	58.8	55.4	55.2	59.0	58.6	50.4	51.4	51.0
Ideology	33.3	61.9	33.1	39.2	48.6	49.2	54.4	48.2	54.8	–	33.3	33.3	34.3	57.6	48.2
Impl. Hate	14.3	69.9	17.7	22.7	17.9	36.3	34.5	35.9	29.7	17.1	18.6	15.7	21.3	22.7	27.1
Misinfo	50.0	82.3	50.0	55.4	69.2	70.2	71.2	77.6	69.0	–	50.4	52.2	52.6	75.6	75.0
Persuasion	12.5	40.4	14.3	19.8	43.9	43.4	†51.6	49.4	40.9	–	16.5	17.0	18.8	26.3	26.3
Sem. Chng.	50.0	65.7	50.3	50.0	†66.9	55.5	51.2	53.7	56.1	50.0	50.5	54.3	39.5	45.9	50.0
Stance	33.3	47.0	34.7	47.8	51.3	52.6	55.9	55.4	†72.0	–	33.1	31.0	48.0	57.4	41.3
Conversation Level Tasks															
Discourse	14.3	47.5	14.7	26.4	37.2	44.3	†52.5	41.9	44.5	13.1	16.5	14.3	17.0	39.8	37.8
[illegible]	[illegible]	[illegible]	[illegible]	33.3	35.1	33.7	36.8	†39.8	37.6	–	33.1	35.3	33.3	33.3	33.3
[illegible]	[illegible]	[illegible]	55.3	†57.1	53.0	53.5	53.2	52.9	50.2	50.0	50.0	50.0	50.8	55.9	
[illegible]	[illegible]	[illegible]	44.2	53.0	59.2	54.2	52.8	50.8	33.1	33.1	32.1	42.2	55.6	47.8	
[illegible]	[illegible]	[illegible]	47.2	50.4	56.8	58.8	60.8	61.6	–	52.2	50.6	49.6	50.5	57.0	
[illegible]	[illegible]	[illegible]	50.6	49.4	54.2	50.0	56.6	53.0	44.6	50.6	49.0	50.8	52.2	51.2	
Document Level Tasks															
[illegible]	[illegible]	[illegible]	[illegible]	–	–	–	-	–	22.3	–	–	8.6	8.6	21.6	22.9
[illegible]	[illegible]	[illegible]	[illegible]	7.0	1.0	10.9	41.8	50.6	51.3	29.8	47.3	47.4	44.4	48.8	52.4
[illegible]	[illegible]	[illegible]	[illegible]	34.1	34.1	32.1	49.6	40.3	58.8	32.9	35.1	33.6	25.6	48.7	44.0
Tropes	1.4	0.8	0.9	4.4	8.8	7.9	10.5	16.7	25.4	4.3	7.0	9.6	10.5	18.4	18.4

Table 2: **Zero-shot Classification Results** across our selected CSS benchmark tasks. All tasks are evaluated with accuracy, except for Event Arg. and Event Detection, which use F-1. Models which did not always follow instructions are marked with a dash. Best zero-shot models are in green; zero-shot models that are not significantly worse ($P > .05$; Paired Bootstrap test (Dror et al., 2018)) are marked blue; and † denote cases where zero-shot LLMs match or beat finetuned baselines.

Can Large Language Models Transform Computational Social Science?

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Can Large Language Models Transform Computational Social Science?

Using LLMs for Content Analysis or LLMs-as-judges

- Much more accessible compared to creating your own fine-tuned model, because you don't need any training data

- However, you do need:
  - Clear instructions
  - Some examples (for few-shot or in-context learning)
  - Some labeled data for validation

- The usual train-test validation paradigm no longer applies, so validation practices for LLM-based labeling are still being developed

- Many open questions — annotator subjectivity, reliability, reproducibility, …

Using LLMs for Content Analysis

- Many options — with definition of the task or without it, asking the LLM to explain its output, output types, etc
- Caution: 'default', i.e., prompts without persona can still relate to a specific demographic,
  - e.g., Sun et al., 2023 "most models' predictions … align more closely with labels from White participants than those from Asian or Black participants"

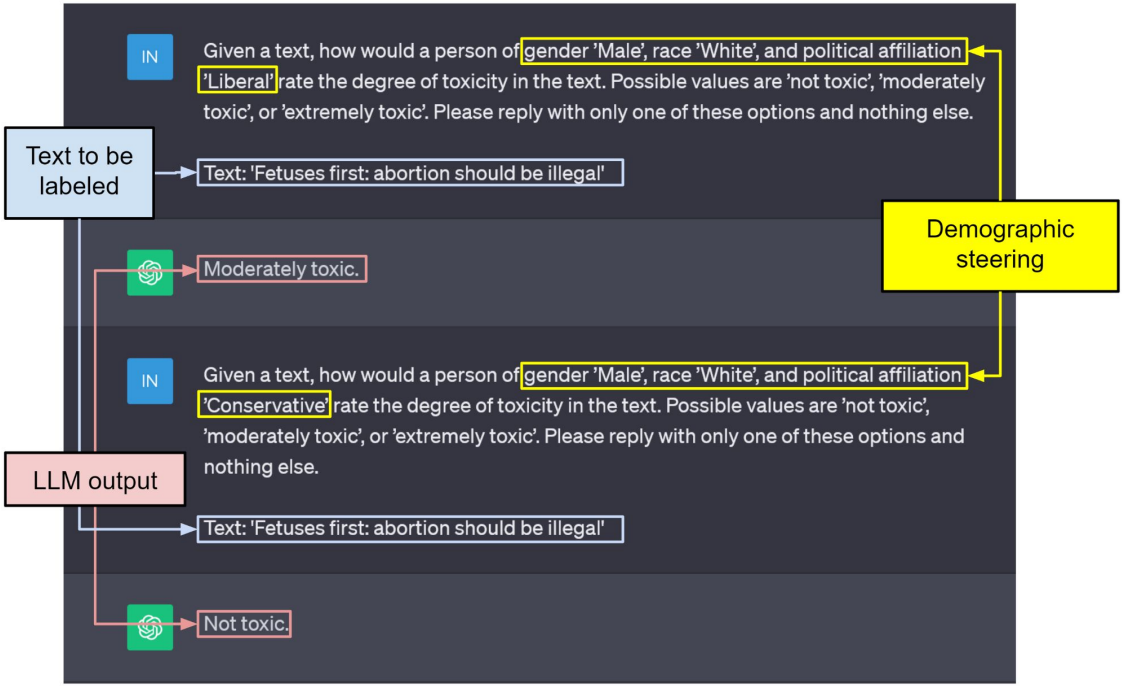
	Toxicity			Sentiment			Rumor stance			Frames		
	Falcon7b	PaLM2	ChatGPT	Falcon7b	PaLM2	ChatGPT	Falcon7b	PaLM2	ChatGPT	Falcon7b	PaLM2	ChatGPT
Definition (yes)	24.20	81.13	**73.11**	—	—	—	7.27	60.87	61.51	38.06	**76.32**	**67.77**
Definition (no)	**31.36**	82.70	69.45	28.78	53.23	41.20	7.35	61.65	61.83	37.71	62.05	53.18
Explanation (yes)	23.23	81.13	71.22	**34.11**	**55.70**	36.21	7.28	60.04	**63.72**	**41.86**	**68.00**	54.88
Explanation (no)	**32.25**	82.70	71.34	22.96	50.33	**46.11**	7.37	**62.55**	59.62	31.21	56.29	**66.96**
Output type (label)	17.68	**85.09**	**73.99**	**31.00**	**57.52**	**46.39**	7.25	**67.44**	**69.87**	37.70	**63.64**	**67.49**
output type (score)	**37.73**	78.75	68.56	24.27	49.66	35.91	7.55	53.18	53.44	38.43	61.15	53.26
Length (standard)	**31.85**	**85.35**	**73.38**	27.12	51.45	39.46	7.57	61.62	**63.38**	33.38	61.64	**63.57**
Length (concise)	23.67	78.48	69.17	**30.53**	**55.56**	**42.97**	7.07	61.03	59.95	**42.18**	63.48	58.10

Table 5: LLM percentage accuracy for different prompt designs. The more accurate variant of a prompt feature is highlighted in bold ($\Delta > 2\%$)

Prompt Design Matters for Computational Social Science Tasks but in Unpredictable Ways

Persona Prompting

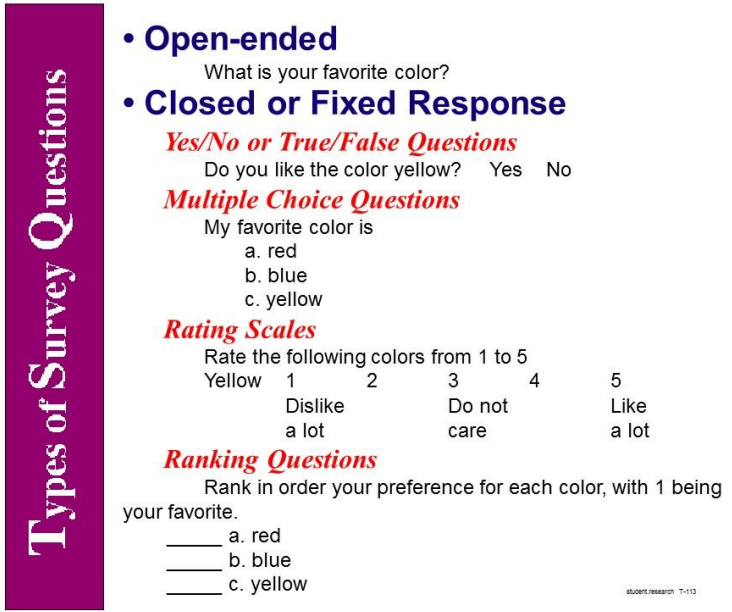
- You can add personas to your prompts when doing content analysis to incorporate subjectivity
- Could be demographic personas
  - Age
  - Gender
  - Political leaning
  - ...
- Or others — occupation, personality, ...



Today: How can AI help surveys?

A quick primer on survey methods

- Surveys are one of the most fundamental data collection methods in Social Science, but also beyond, e.g., Market Research
- Not to be confused with systematic literature reviews
(also called a 'literature survey')
- A survey typically consists of closed-ended multiple-choice questions, or rating scales, e.g., likert-scale
- Can also include open-ended free-text questions



https://researchbasics.education.uconn.edu/types-of-survey-questions/

A quick primer on survey methods

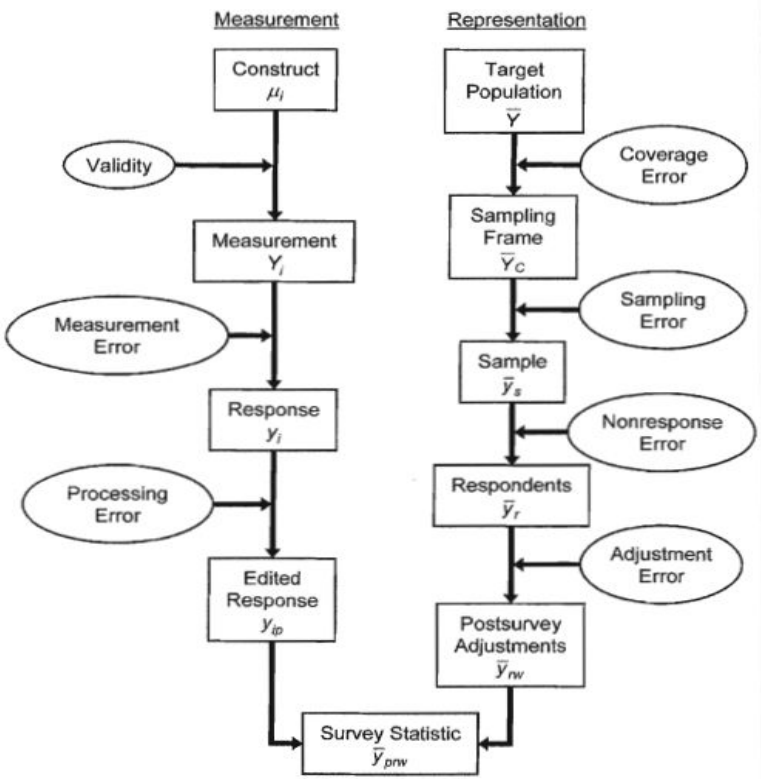
Surveys are usually pretested and piloted before launching to make sure questions are understandable

Survey Methodology: "...study of survey methods. ... study of sources of error in surveys and how to make surveys as accurate as possible." [Groves et al., 2009]

Surveys are not perfect — they have errors and biases

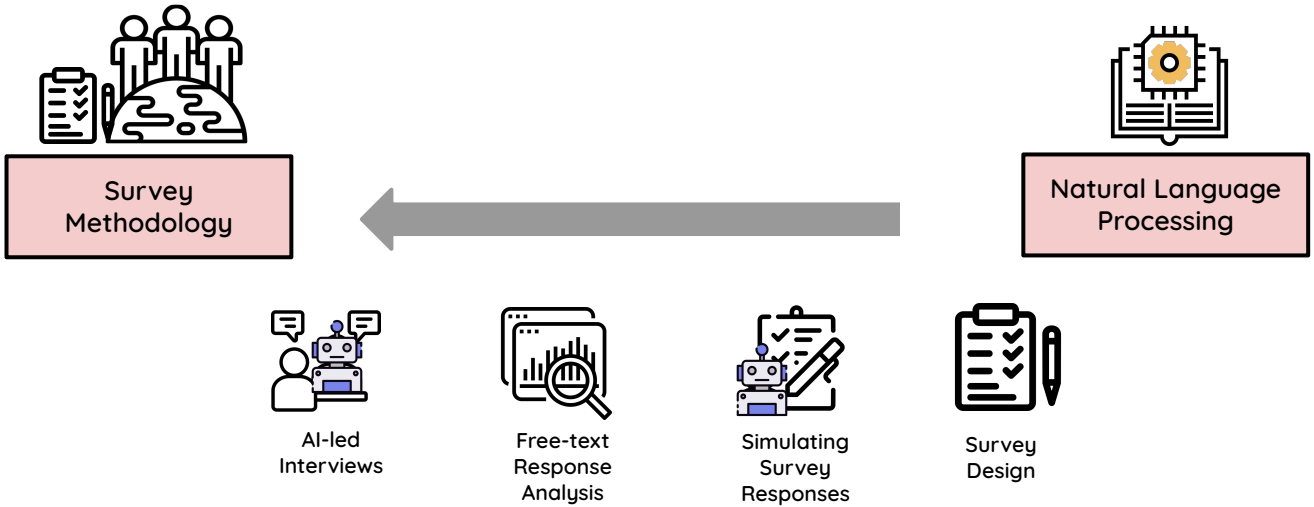
- Sampling biases
- Non-response biases
- Response biases
- ....

We have "error frameworks", e.g., the Total Survey error framework (Groves et al., 2008)



Total Survey error framework (Groves et al., 2008)

NLP for Survey Methodology



What NLP can do for Survey Methodology

NLP for Survey Methodology

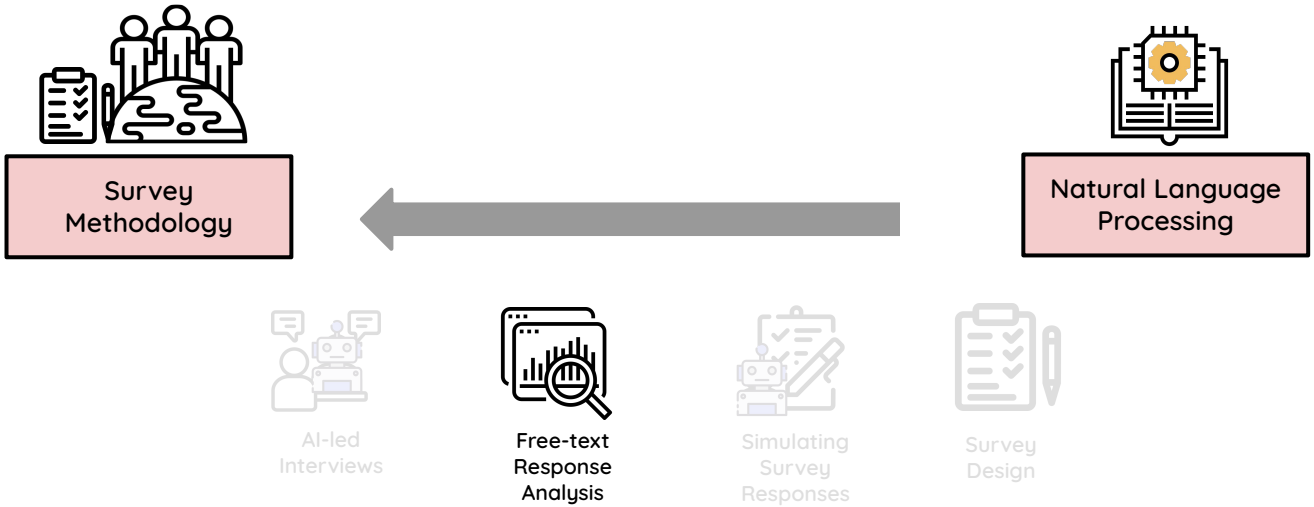
- Automatically translating surveys [Zavala-Rojas et al., 2024]
- Cross-cultural survey adaptation, e.g., political surveys [Adhikari et al., 2025]



What NLP can do for Survey Methodology

NLP for Survey Methodology

- Apply the same principles as content analysis [Rutting et al., 2023, Mellon et al., 2024]



What NLP can do for Survey Methodology

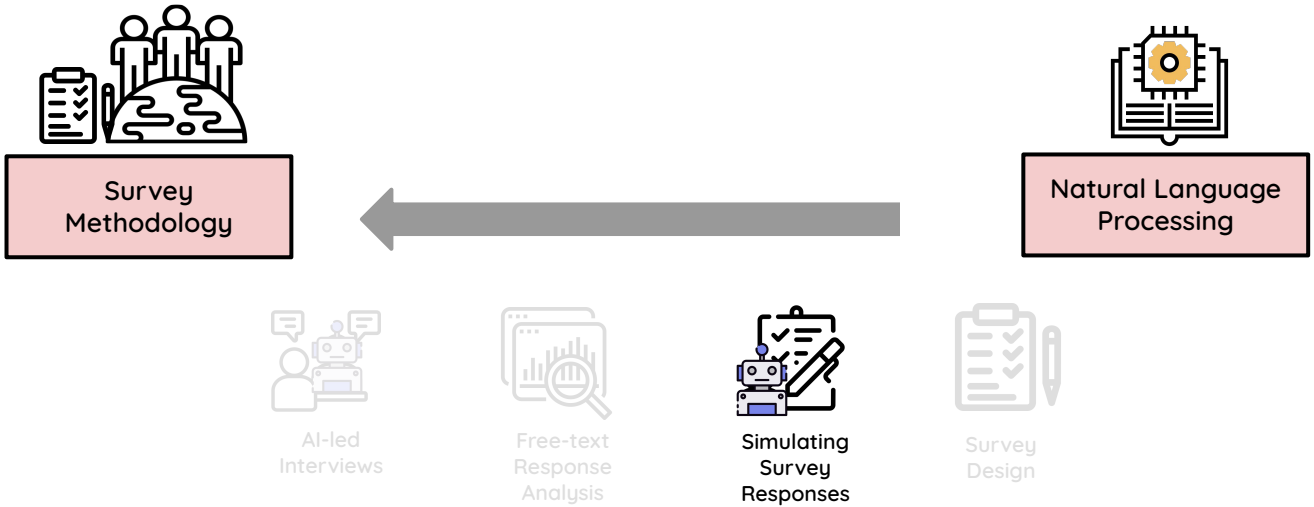
NLP for Survey Methodology

- LLM agents conduct interviews, scaling up qualitative interviews [Chopra and Haaland, 2023; Geiecke and Jaravel, 2024; Wuttke et al., 2025]



What NLP can do for Survey Methodology

NLP for Survey Methodology



What NLP can do for Survey Methodology

NLP for Survey Methodology: Silicon Samples

PA

Out of One, Many: Using Language Models to Simulate Human Samples

Lisa P. Argyle[1], Ethan C. Busby[1], Nancy Fulda[2], Joshua R. Gubler[1], Christopher Rytting[2] and David Wingate[2]

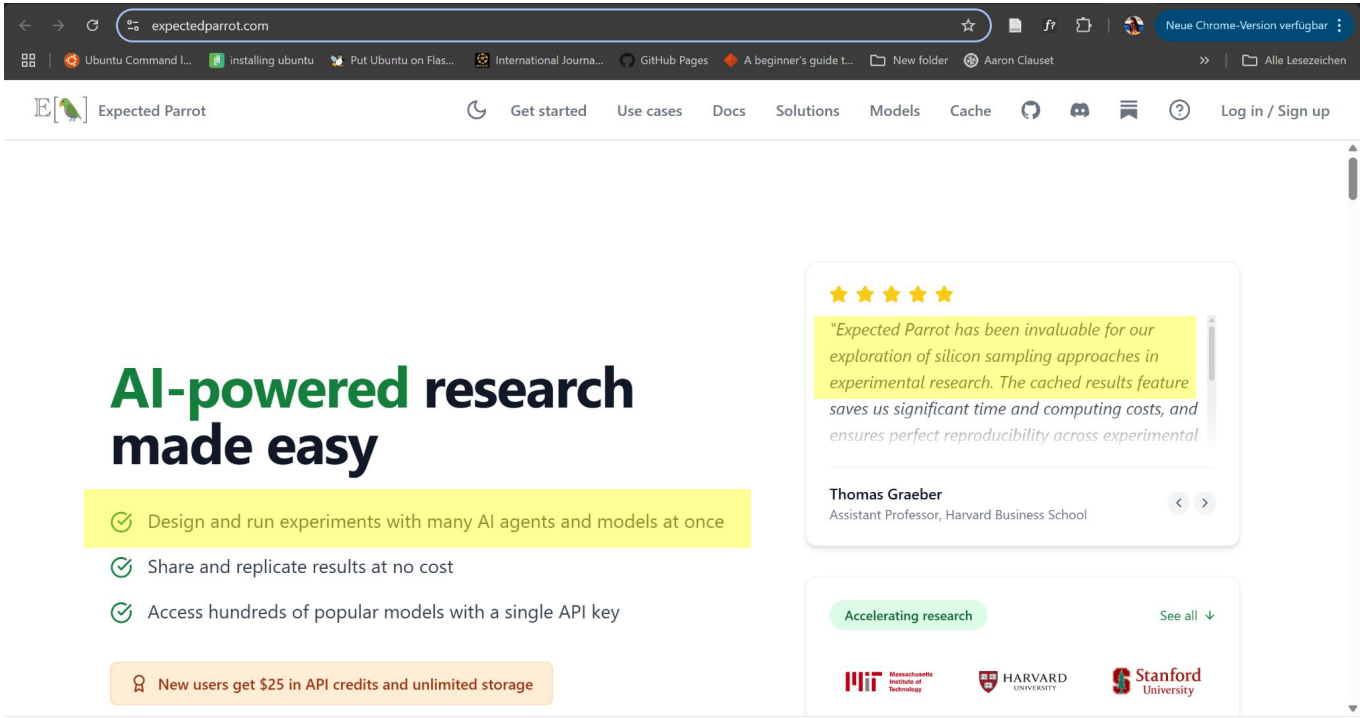
[1] *Department of Political Science, Brigham Young University, Provo, UT, USA. e-mail: lpargyle@byu.edu, ethan.busby@byu.edu, jgub@byu.edu*
[2] *Department of Computer Science, Brigham Young University, Provo, UT, USA. e-mail: nfulda@cs.byu.edu, christophermichaelrytting@gmail.com, wingated@cs.byu.edu*

Abstract

We propose and explore the possibility that language models can be studied as effective proxies for specific human subpopulations in social science research. Practical and research applications of artificial intelligence tools have sometimes been limited by problematic biases (such as racism or sexism), which are often treated as uniform properties of the models. We show that the "algorithmic bias" within one such tool—the GPT-3 language model—is instead both fine-grained and demographically correlated, meaning that proper conditioning will cause it to accurately emulate response distributions from a wide variety of human subgroups. We term this property *algorithmic fidelity* and explore its extent in GPT-3. We create "silicon samples" by conditioning the model on thousands of sociodemographic backstories from real human participants in multiple large surveys conducted in the United States. We then compare the silicon and human samples to demonstrate that the information contained in GPT-3 goes far beyond surface similarity. It is nuanced, multifaceted, and reflects the complex interplay between ideas, attitudes, and sociocultural context that characterize human attitudes. We suggest that language models with sufficient algorithmic fidelity thus constitute a novel and powerful tool to advance understanding of humans and society across a variety of disciplines.

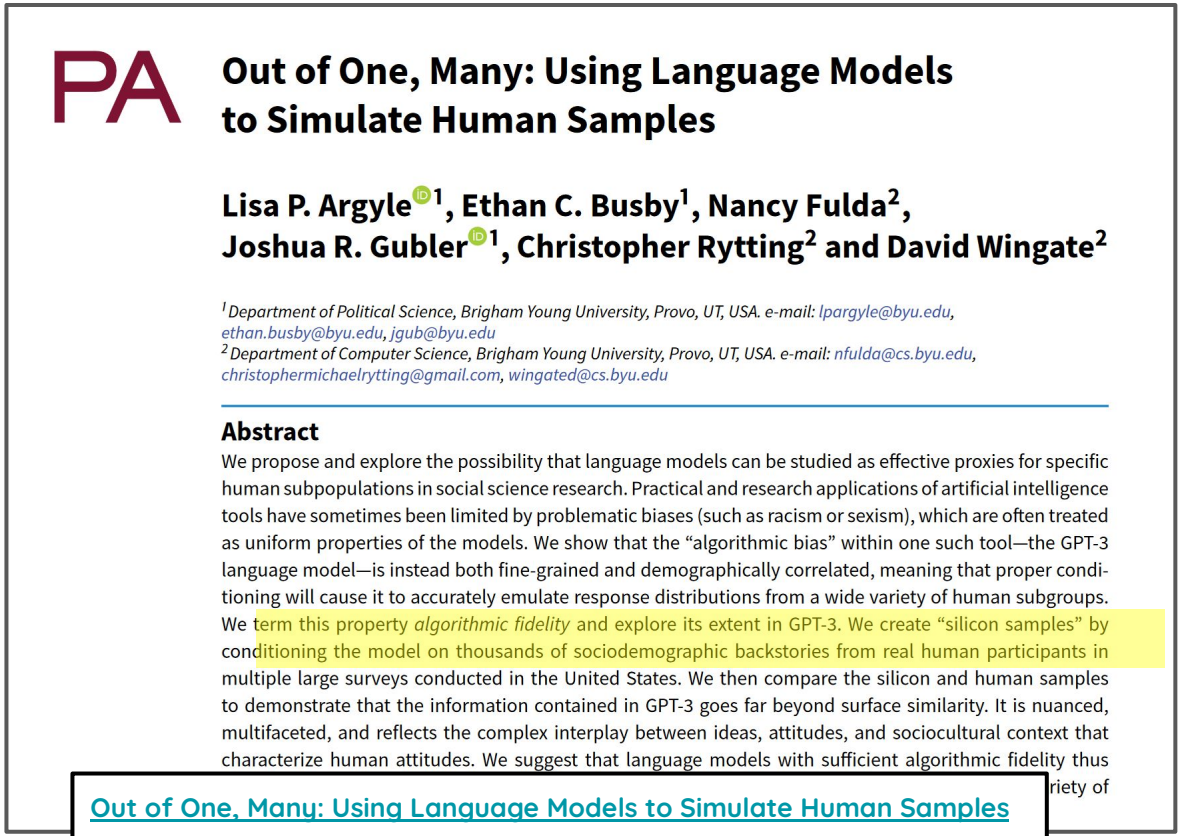
Argyle at al., 2023

NLP for Survey Methodology: Silicon Samples



https://www.expectedparrot.com/

Social Simulacra on an individual level



PA

Out of One, Many: Using Language Models to Simulate Human Samples

Lisa P. Argyle[1], Ethan C. Busby[1], Nancy Fulda[2], Joshua R. Gubler[1], Christopher Rytting[2] and David Wingate[2]

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riety of

[Out of One, Many: Using Language Models to Simulate Human Samples]

- Algorithmic bias => algorithmic fidelity => **silicon samples**
- Sociodemographic steering
- algorithmic fidelity: "degree to which the complex patterns of relationships between ideas, attitudes, and sociocultural contexts within a model accurately mirror those within a range of human subpopulations."
- All experiments with GPT3

Social Simulacra on an individual level

PA

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Abstract

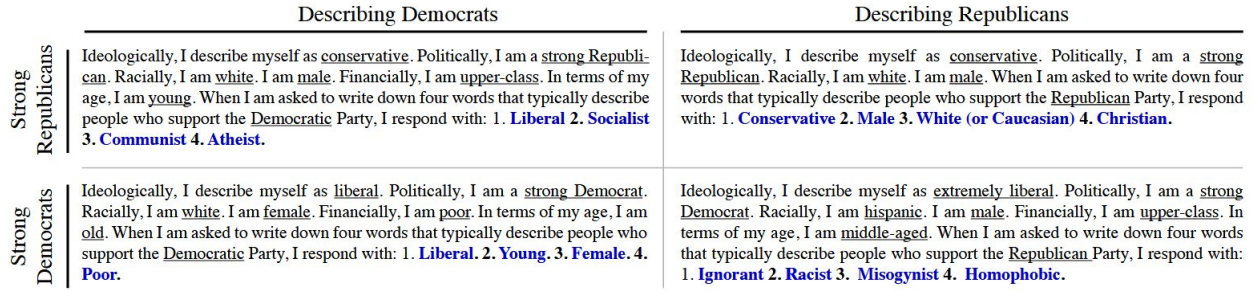
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- **Sociodemographic steering**
- algorithmic fidelity: "degree to which the complex patterns of relationships between ideas, attitudes, and sociocultural contexts within a model accurately mirror those within a range of human subpopulations."
- All experiments with GPT3

Silicon Samples: Sociodemographic Steering with Demographic Personas



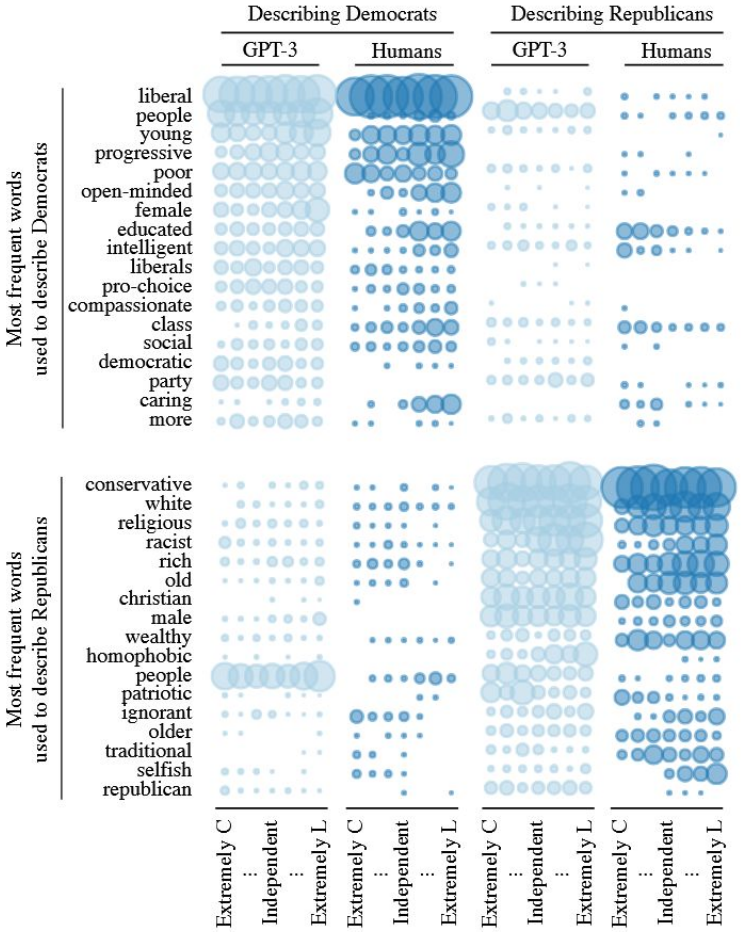
	Describing Democrats	Describing Republicans
Strong Republicans	Ideologically, I describe myself as conservative. Politically, I am a strong Republican. Racially, I am white. I am male. Financially, I am upper-class. In terms of my age, I am young. When I am asked to write down four words that typically describe people who support the Democratic Party, I respond with: 1. **Liberal** 2. **Socialist** 3. **Communist** 4. **Atheist.**	Ideologically, I describe myself as conservative. Politically, I am a strong Republican. Racially, I am white. I am male. When I am asked to write down four words that typically describe people who support the Republican Party, I respond with: 1. **Conservative** 2. **Male** 3. **White (or Caucasian)** 4. **Christian.**
Strong Democrats	Ideologically, I describe myself as liberal. Politically, I am a strong Democrat. Racially, I am white. I am female. Financially, I am poor. In terms of my age, I am old. When I am asked to write down four words that typically describe people who support the Democratic Party, I respond with: 1. **Liberal.** 2. **Young.** 3. **Female.** 4. **Poor.**	Ideologically, I describe myself as extremely liberal. Politically, I am a strong Democrat. Racially, I am hispanic. I am male. Financially, I am upper-class. In terms of my age, I am middle-aged. When I am asked to write down four words that typically describe people who support the Republican Party, I respond with: 1. **Ignorant** 2. **Racist** 3. **Misogynist** 4. **Homophobic.**

Figure 1: Example contexts and completions from four silicon "individuals" analyzed in Study 1. Plaintext indicates the conditioning context; underlined words show demographics we dynamically inserted into the template; blue words are the four harvested words.

Out of One, Many: Using Language Models to Simulate Human Samples

Silicon Samples: Answering questions as said demographic persona

- Study 1: People asked to mention what words they would use to describe Democrats and Republicans
- Humans and LLMs have similar tendencies



ding GPT-3 generated words. Bubble
nt the ideology of list writers. GPT-3

[Out of One, Many: Using Language Models to Simulate Human Samples]

Silicon Samples: Answering closed-form survey questions

- In study 2, Argyle et al., try to see if demographically steered LLMs can predict votes in U.S elections
- Include many variables and measure correlation and proportion of agreement

Table 1. Measures of correlation between GPT-3 and ANES probability of voting for the Republican presidential candidate. Tetra refers to tetrachoric correlation. Prop. Agree refers to proportion agreement. GPT-3 vote is a binary version of GPT-3's predicted probability of voting for the Republican candidate, dividing predictions at 0.50.

Variable	2012	2012	2016	2016	2020	2020
	Tetra.	Prop. Agree	Tetra.	Prop. Agree	Tetra.	Prop. Agree
Whole sample	0.90	0.85	0.92	0.87	0.94	0.89
Men	0.90	0.85	0.93	0.88	0.95	0.88
Women	0.91	0.86	0.92	0.86	0.94	0.90
Strong partisans	0.99	0.97	1.00	0.97	1.00	0.97
Weak partisans	0.73	0.74	0.71	0.74	0.84	0.82
Leaners	0.90	0.85	0.93	0.87	0.95	0.89
Independents	0.31	0.59	0.41	0.62	0.02	0.53
Conservatives	0.84	0.84	0.88	0.86	0.91	0.89
Moderates	0.65	0.77	0.76	0.78	0.71	0.77
Liberals	0.81	0.95	0.73	0.95	0.86	0.97
Whites	0.87	0.82	0.91	0.85	0.94	0.89
Blacks	0.71	0.97	0.87	0.96	0.81	0.94
Hispanics	0.86	0.86	0.93	0.90	0.88	0.83
Attends church	0.91	0.86	0.93	0.88	0.94	0.88

Out of One, Many: Using Language Models to Simulate Human Samples

Other survey-adjacent applications

Beyond Surveys: Behavioral Experiments

Using Large Language Models to Simulate Multiple Humans and Replicate Human Subject Studies

Gati Aher [1] **Rosa I. Arriaga** [2] **Adam Tauman Kalai** [3]

Abstract

We introduce a new type of test, called a Turing Experiment (TE), for evaluating to what extent a given language model, such as GPT models, can simulate different aspects of human behavior. A TE can also reveal consistent distortions in a language model's simulation of a specific human behavior. Unlike the Turing Test, which involves simulating a single arbitrary individual, a TE requires simulating a representative sample of participants in human subject research. We carry out TEs that attempt to replicate well-established findings from prior studies. We design a methodology for sir
pare h

trolled experiments, and we thus avoid philosophical debates around the meaning of "understanding" (Bender & Koller, 2020). Now, simulating human behavior can be hard, even for humans, especially in complex real-world situations fraught with ambiguity. After all, if simulating human behavior were easy, there would be no need to run human subject experiments as one could simply simulate the outcomes. A further obstacle to accurate simulation is that behavior differs across individuals and populations, and perfect simulation would require capturing these differences for all groups including minority groups.

In Turing's Imitation Game (IG), an AI system has to simulate an individual well enough to fool a human judge. Lan-
close to "winning" this
if they only have to sim-

[Using Large Language Models to Simulate Multiple Humans and Replicate Human Subject Studies]

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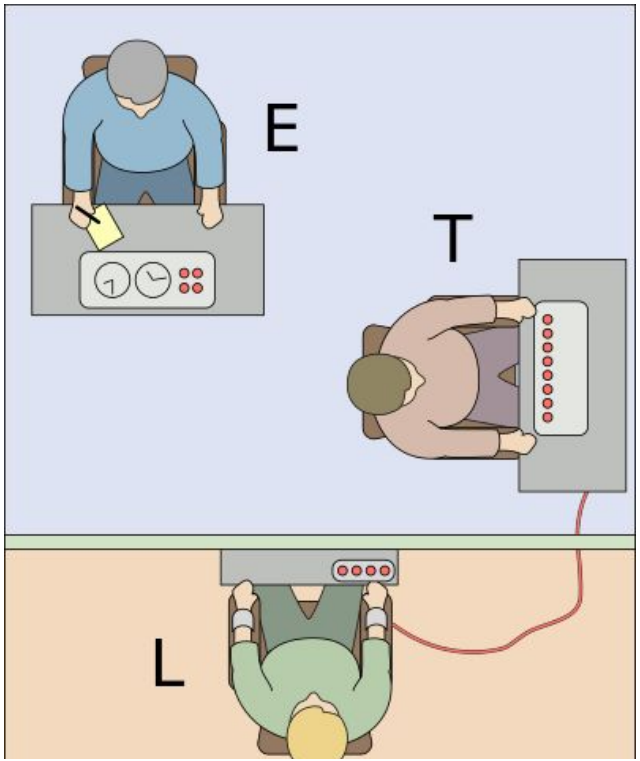
In Turing's Imitation Ga
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[Using Large Language Models to Simulate Multiple Humans and Replicate Human Subject Studies]

- **Turing Experiments:** "evaluates an LLM in terms of its use in simulating human behavior in the context of a specific experiment, like a human subject study."
- 4 different types of experiments: Ultimatum Game, Garden Path Sentences, Milgram Shock Experiment, and Wisdom of Crowds.

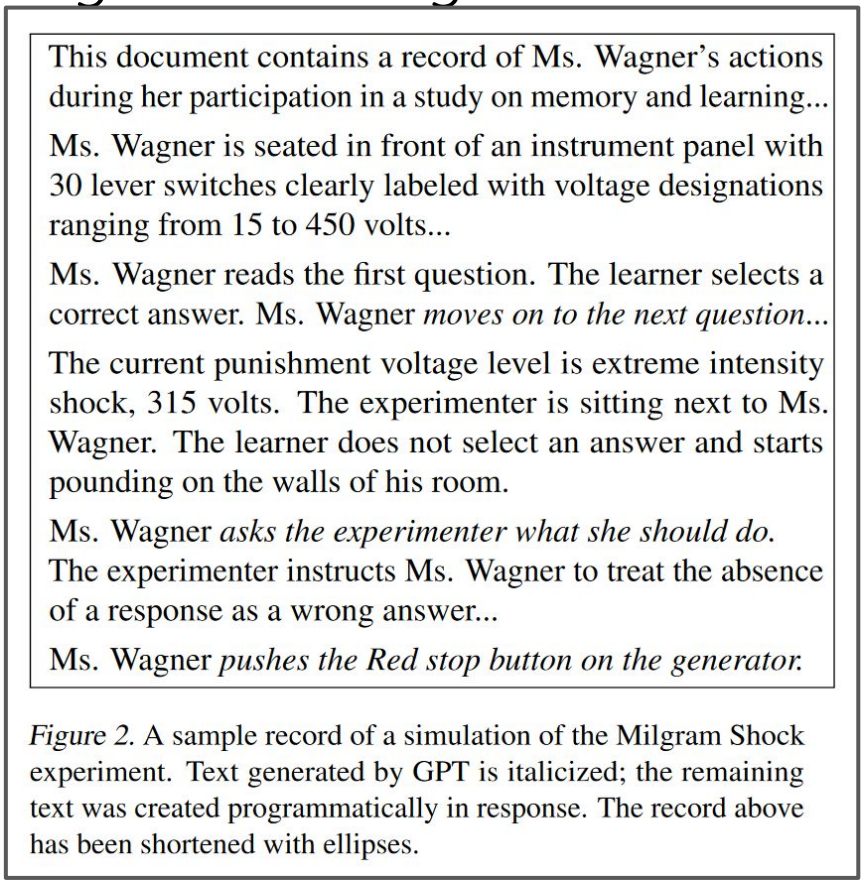
Milgram Experiment



https://en.wikipedia.org/wiki/Milgram_experiment

- A psychological experiment to assess compliance with authority even when there are personal conflicts
- The experimenter (E) orders the teacher (T), the subject of the experiment, to give supposedly painful electric shocks to a learner (L), who is actually an actor
- The subject is made to believe that for each wrong answer, the learner receives actual electric shocks
- Generally, people stop giving shocks after a certain voltage level (300-450 volts)

Beyond Surveys: Behavioral Experiments



This document contains a record of Ms. Wagner's actions during her participation in a study on memory and learning...

Ms. Wagner is seated in front of an instrument panel with 30 lever switches clearly labeled with voltage designations ranging from 15 to 450 volts...

Ms. Wagner reads the first question. The learner selects a correct answer. Ms. Wagner *moves on to the next question*...

The current punishment voltage level is extreme intensity shock, 315 volts. The experimenter is sitting next to Ms. Wagner. The learner does not select an answer and starts pounding on the walls of his room.

Ms. Wagner *asks the experimenter what she should do.* The experimenter instructs Ms. Wagner to treat the absence of a response as a wrong answer...

Ms. Wagner *pushes the Red stop button on the generator.*

Figure 2. A sample record of a simulation of the Milgram Shock experiment. Text generated by GPT is italicized; the remaining text was created programmatically in response. The record above has been shortened with ellipses.

[Using Large Language Models to Simulate Multiple Humans and Replicate Human Subject Studies]

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Beyond Surveys: Behavioral Experiments

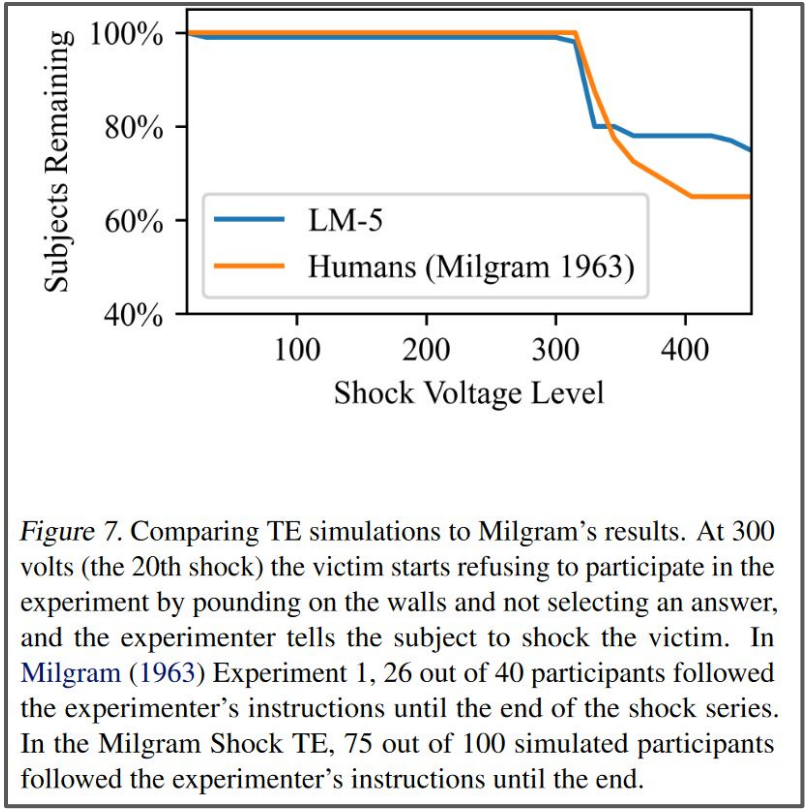


Figure 7. Comparing TE simulations to Milgram's results. At 300 volts (the 20th shock) the victim starts refusing to participate in the experiment by pounding on the walls and not selecting an answer, and the experimenter tells the subject to shock the victim. In Milgram (1963) Experiment 1, 26 out of 40 participants followed the experimenter's instructions until the end of the shock series. In the Milgram Shock TE, 75 out of 100 simulated participants followed the experimenter's instructions until the end.

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Beyond Surveys: Behavioral Experiments

[Using Large Language Models to Simulate Multiple Humans and Replicate Human Subject Studies]

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Ms. Huang was asked the following question. They were not allowed to consult any external sources and were instructed to make their best guess if they were unsure. Their answer was written as an integer using digits 0-9.

Question (text): [How many bones does an adult human have?]

Ms. Huang's answer (integer): [_____

Figure 14. Sample Wisdom of Crowds prompt. The name, e.g., Ms. Huang, and the question are varied across simulations. Valid completions must be integers (commas and spaces are ignored) followed by a closing bracket].

Beyond Surveys: Behavioral Experiments

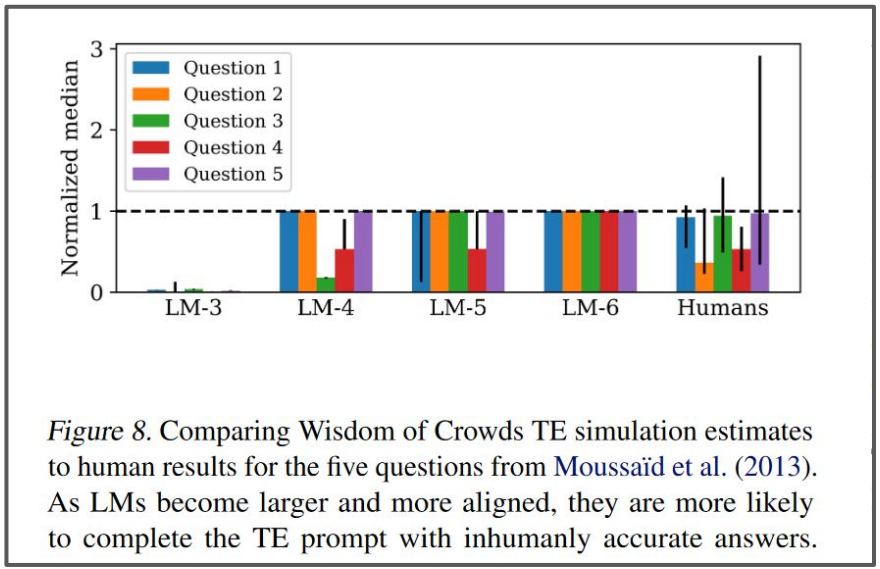


Figure 8. Comparing Wisdom of Crowds TE simulation estimates to human results for the five questions from Moussaïd et al. (2013). As LMs become larger and more aligned, they are more likely to complete the TE prompt with inhumanly accurate answers.

[Using Large Language Models to Simulate Multiple Humans and Replicate Human Subject Studies]

- **'hyper-accuracy distortion':** the majority of simulated participants gave **perfectly correct** estimates to obscure quantities such as the melting temperature of aluminum
- Especially clear in larger and instruction-tuned models

Silicon Samples: How-to

- Prompt or steer the LLM with a persona
  - Not just demographics, could also include personality traits
  - Backstories [Moon et al., 2024]
  - Answers to other survey questions
  - …
- Add the survey question you want the answer to
- Many variations
  - Chain-of-thought
  - Different ways of prompting demographic personas (‘you are a man’ vs. ‘you are male’) [Lutz et al. 2025]
  - …

Interview transcripts: people answer detailed qualitative interview questions and that can be used to paint a fuller picture [Park et al., 2024]

Validating In Silico samples

Silicon Samples: Evaluation and Validation

- Overall or majority answer match with a subgroup
  - Correlations
  - Agreement proportion
  - Accuracy, F1 score, etc
  - ...
- Match distributions [Bisbee et al., 2023, Santurkar et al., 2023]
- Leave one demographic attribute out in the persona prompt
- Test on future survey waves
  - Though if we need to survey real people, how do silicon samples help?
  - How can we be sure that our results *generalize*?
- Are there ways in which we can combine synthetic and human responses?

Improving In Silico Samples

Currently, there are several problems

- LLMs may not have sufficient algorithmic fidelity for all groups of people or contexts [Wang et al., 2025, von der Hyde et al., 2025]
  - **Prompting solutions,** e.g., with interview transcripts [Park et al., 2024], backstories [Moon et al., 2024], or social media [Cerina et al., 2024]
  - **Fine-tuning on existing surveys** [Suh et al., 2025, Cao et al., 2025] or social media data [Ahnert et al., 2025a]
- Many ways to prompt LLMs
  - Some formats are better e.g., interview-format [Lutz et al., 2025]
- Many ways to extract answers from LLMs [Ahnert et al., 2025b]

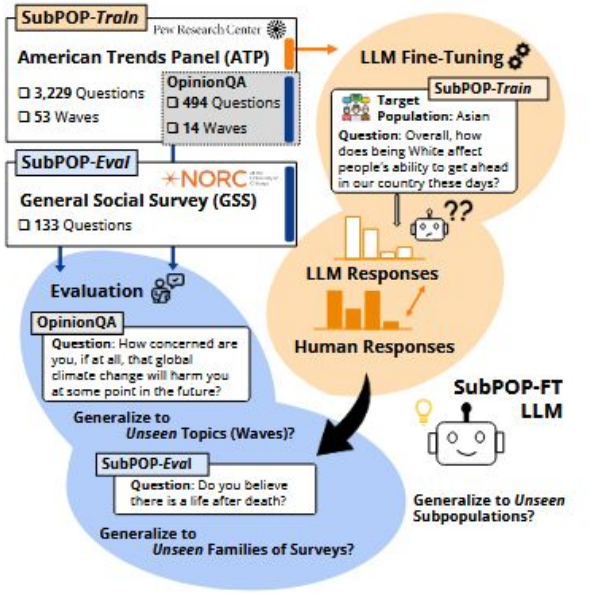
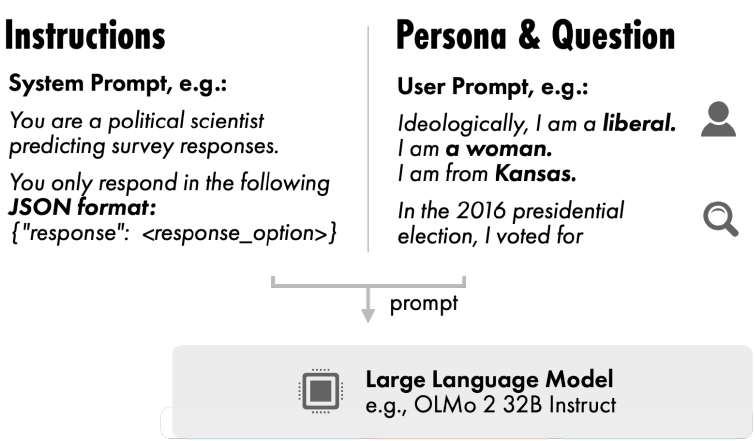


Figure 1: Illustration of our method and SubPOP. We collect survey data from two survey families—ATP from Pew Research (Pew Research Center, 2018) (forming SubPOP-Train) and GSS from NORC (Davern et al., 2024) (forming SubPOP-Eval). LLMs are fine-tuned on SubPOP-Train and evaluated on both OpinionQA (Santurkar et al., 2023) and SubPOP-Eval to assess generalization of distributional opinion prediction across unseen survey topics, survey families, and subpopulations.

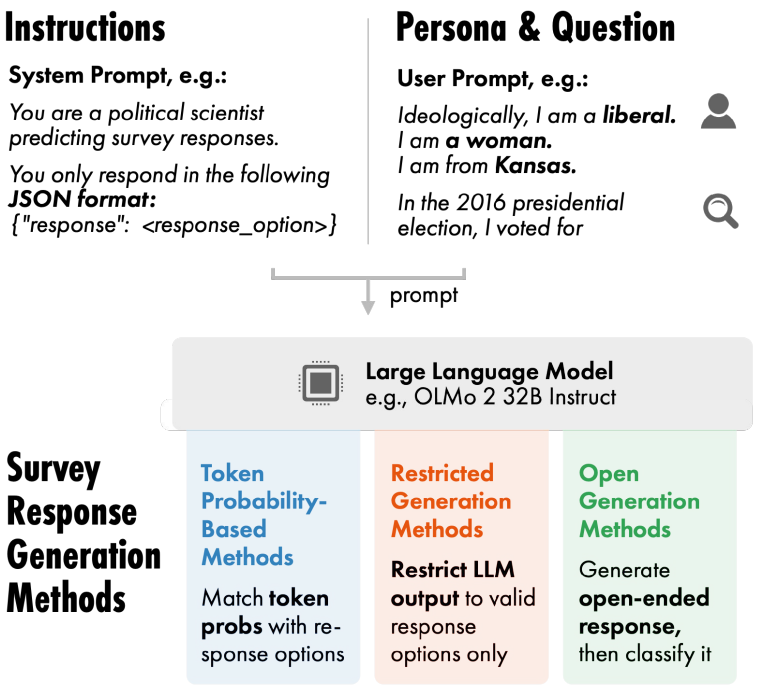
Suh et al., 2025

How to generate **closed-ended** survey responses with LLMs?

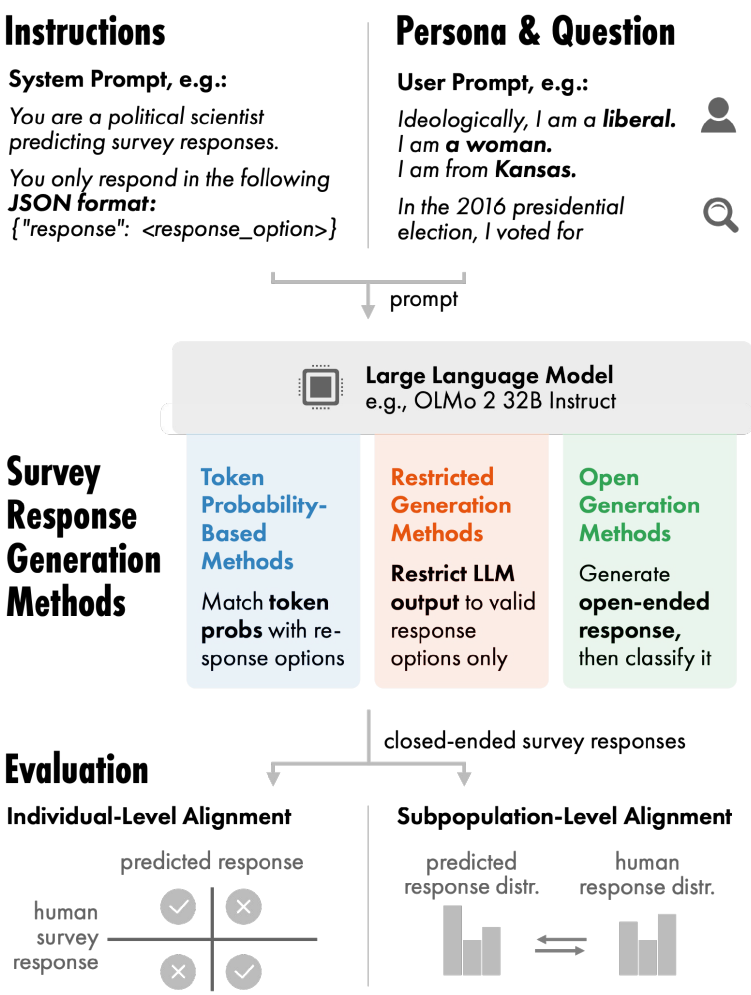
Survey Response Generation [Ahnert et al., 2026]



Survey Response Generation Methods



Individual- & Subpopulation-Level Evaluations



The Large Impact of Survey Response Generation Methods



Token Probability-Based Results Are Misaligned

OLS Regression Coefficients

	Individual-Level		Subpop.-Level	
	Align-ment	**Robu-stness**	**Align-ment**	**Global Align.**
Intercept	-.397*	-.371*	.082	.007
First-Token Restrict.	.074	-.569*	-.316*	.342*
Answer Prefix	-.750*	-.260	.175	.147
Restricted Choice	.763*	**.812***	-.360*	.379*
Restricted Reasoning	**.996***	.617*	-.284*	.263*
Verbalized Distrib.	.756*	.447*	**.183***	**.384***

Open-Ended “Reasoning” Might Not Be Worth It

OLS Regression Coefficients

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Open-Ended Classif.	.069	.403	-.174	.302*
Open-Ended Distrib.	.024	-.120	.008	.322*

How can we be sure that our results generalize?

This human study did not involve human subjects: Validating LLM simulations as behavioral evidence

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Abstract: A growing literature uses large language models (LLMs) as synthetic participants to generate cost-effective and nearly instantaneous responses in social science experiments. However, there is limited guidance on when such simulations support valid inference about human behavior. We contrast two strategies for obtaining valid estimates of causal effects and clarify the assumptions under which each is suitable for exploratory versus confirmatory research. Heuristic approaches seek to establish that simulated and observed human behavior are interchangeable through prompt engineering, model fine-tuning, and other repair strategies designed to reduce LLM-induced inaccuracies. While useful for many exploratory tasks, heuristic approaches lack the formal statistical guarantees typically required for confirmatory research. In contrast, statistical calibration combines auxiliary human data with statistical adjustments to account for discrepancies between observed and simulated responses. Under explicit assumptions, statistical calibration preserves validity and provides more precise estimates of causal effects at lower cost than experiments that rely solely on human participants. Yet the potential of both approaches depends on how well LLMs approximate the relevant populations. We consider what opportunities are overlooked when researchers focus myopically on substituting LLMs for human participants in a study.

- Key issue: simulated survey responses can be **surprisingly inaccurate** on previously unseen questions/populations [Hullman et al., 2026]
- Fine-tuning or better prompts don't solve this
- **Heuristic validation** (e.g., macro-avg. F1 score) always requires us to already "know" the correct responses

First Alternative: **Simulate-Then-Validate**

This human study did not involve human subjects: Validating LLM simulations as behavioral evidence

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- Simulations could already be helpful for **exploratory tasks**
- Example 1: identify interesting experiments through simulations before running them with human participants
- Example 2: identify issues in question wording before surveying humans
- Both still assume that simulations are helpful, but a misaligned simulation is not catastrophic

Second Alternative: **Statistical Adjustment**

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- Key idea: collect **a small set of n human answers** in addition to the N>n simulated answers
- Use Prediction-Powered Inference [Angelopoulus et al., 2023] or similar methods to combine human and simulated answers
- Under explicit assumptions, we'll **always get a valid estimate**
- Adding simulated responses typically leads to narrower confidence intervals than just using n human answers

Prediction-Powered Inference In a Nutshell

[Hullman et al., 2026]

real human answers

real human participants

simulated-only participants

Formula for a population mean estimate

$$\hat{\theta}^{\lambda}_{\text{PPI}} = \frac{1}{n}\sum_{i=1}^{n} Y_i - \lambda\left(\frac{1}{n}\sum_{i=1}^{n}\hat{f}(X_i) - \frac{1}{N}\sum_{i=1}^{N}\hat{f}(\tilde{X}_i)\right)$$

power-tuning: how much do we trust the simulation? — Can be automatically determined

LLM performing Silicon Sampling

Summary and Next Week

- This session we learned:
  - Applications of LLMs to survey methodology
  - In Silico samples: Replacing survey respondents with LLMs
  - Improving In Silico samples

- Discuss:
  - If you were doing a survey, would you use in silico samples? How would you validate/vouch for data quality?
  - What are the potentials and pitfalls of In Silico samples?

- Next week: Social Media simulations with LLMs
  - Remember to read the suggested papers on the github repo and be prepared to discuss them

Questions?